

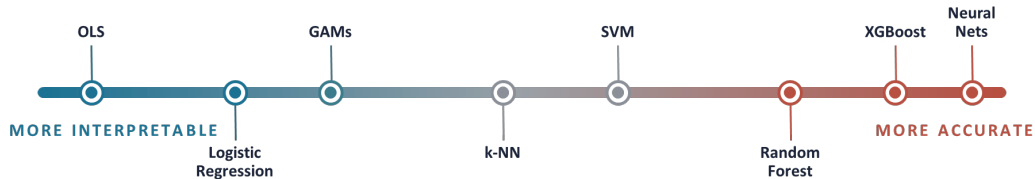
Method-Agnostic Interpretation via Average Marginal Effects

Regression-style inference for any supervised model

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The accuracy–interpretability tradeoff



Is the tradeoff really fundamental?

A familiar coefficient table

	Logistic				XGBoost			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
<i>Panel A: Demographics</i>								
age	0.0060*** (0.0001)	0.0060*** (0.0001)	0.0024*** (0.0001)	0.0027*** (0.0001)	0.0059*** (0.0002)	0.0059*** (0.0001)	0.0029*** (0.0001)	0.0031*** (0.0001)
female	-0.1730*** (0.0034)	-0.1648*** (0.0034)	-0.0275*** (0.0041)	-0.0269*** (0.0041)	-0.1592*** (0.0032)	-0.1446*** (0.0032)	-0.0138*** (0.0028)	-0.0122*** (0.0032)
education_num	0.0523*** (0.0006)	0.0393*** (0.0007)	0.0400*** (0.0006)	0.0306*** (0.0006)	0.0610*** (0.0023)	0.0452*** (0.0017)	0.0405*** (0.0011)	0.0312*** (0.0011)
<i>Panel B: Work Characteristics</i>								
hours_per_week	—	0.0048*** (0.0001)	—	0.0033*** (0.0001)	—	0.0084*** (0.0028)	—	0.0032 (0.0025)
government	—	0.0029 (0.0051)	—	0.0033 (0.0045)	—	-0.0067* (0.0037)	—	-0.0003 (0.0026)
self_employed	—	-0.0077 (0.0050)	—	-0.0268*** (0.0044)	—	0.0113** (0.0054)	—	-0.0055* (0.0033)
white_collar	—	0.0666*** (0.0056)	—	0.0541*** (0.0051)	—	0.0466*** (0.0026)	—	0.0474*** (0.0017)
blue_collar	—	-0.0264*** (0.0052)	—	-0.0270*** (0.0048)	—	-0.0446*** (0.0041)	—	-0.0251*** (0.0034)
service	—	-0.0697*** (0.0069)	—	-0.0427*** (0.0066)	—	-0.0807*** (0.0049)	—	-0.0343*** (0.0058)
<i>Panel C: Financial</i>								
capital_gain	—	—	0.0000*** (0.0000)	0.0000*** (0.0000)	—	—	-0.0001*** (0.0000)	-0.0001*** (0.0000)
capital_loss	—	—	0.0001*** (0.0000)	0.0001*** (0.0000)	—	—	-0.0001*** (0.0000)	-0.0000*** (0.0000)
married	—	—	0.2854*** (0.0042)	0.2752*** (0.0041)	—	—	0.2383*** (0.0035)	0.2357*** (0.0036)
McFadden R^2	0.199	0.233	0.376	0.400	0.257	0.296	0.467	0.492
Accuracy	0.797	0.806	0.841	0.847	0.805	0.818	0.864	0.872

Notes: Average marginal effects on $P(\text{Income} > \$50k)$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Bootstrap standard errors (200 replicates) in parentheses.

The right four columns are XGBoost.

Same table you already read — estimates, standard errors, significance — for a black-box model.

What this is

- ▶ A method for reading **any** machine learning model as a regression-style coefficient table — estimates, standard errors, significance.
- ▶ Average marginal effects: the expected change in predicted output for a one-unit change in each feature, averaged across all observations.
- ▶ Standard errors obtained via **nonparametric bootstrap** — no distributional assumptions required.
- ▶ Distilled from a full-length companion manuscript (105 pages, complete proofs); a condensed 10-page version is the conference submission.